

Department Overview

- **Department Name:** Department of Chemical Engineering
- **Faculty:** Faculty of Engineering & Architecture
- **Accreditation:** Fully accredited by the **Pakistan Engineering Council (PEC)**.
- **Core Engineering Focus:** Focusing on the transformation of raw feedstock into valuable commercial products (such as transport fuels, agro-industry fertilizers, processed foods, clean water, and construction cement). The curriculum encompasses chemical process and equipment designs, cost estimation, process control, optimization, and the safe scaling of pilot projects to industrial operations.
- **History:** Founded in September 2005 with only 4 faculty members and 14 students at the Takatu Campus. Today, the department grows with 21 faculty members, most of whom are foreign-qualified.
- **Research Laboratories & Domains:** Equipped with state-of-the-art labs actively engaged in cutting-edge research covering Nanotechnology, Algal cultivation, Biomass and bioenergy, Waste management for product recovery, Wastewater treatment, and Polymer composites.

Vision and Mission

- **Vision:** To be among the leading Chemical Engineering Department of the country with programs providing quality graduates for the chemical and allied industry with focus on the research and development.
- **Mission:** The Department of Chemical Engineering is committed to establish conducive environment for the professional education and research in the field of Chemical Engineering addressing the needs of the country and region.

Program Educational Objectives (PEOs)

The department targets producing chemical engineers who can display the following traits within a few years of graduation:

1. Professional competency in chemical engineering to address the needs of the chemical and allied industries, research and development sector.
2. Contribute to the solution of industrial problems, technological and societal issues.
3. Pursue professional development and technical achievement through continuing education.

1. Bachelor of Science in Chemical Engineering (BS Chemical Engineering)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering) from a recognized board or an

equivalent certification with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of **60% marks**.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 45 – 49
- **Minimum Academic Standing:** CGPA $\geq$ 2.0

Curriculum Plans (Program Schema by Semester)

The department maps out two distinct course pathways based on enrollment intake years.

Note: Credit hour structures denote (Theory + Lab).

Track A: For Student Intake Terms Fall-2019 to Fall-2021

- **1st Semester:** Islamic Studies/Ethics (2+0), Pakistan Studies (2+0), Introduction to Computers (1+1), Engineering Drawing (0+1), Calculus & Analytical Geometry (3+0), Introduction to Chemical Engineering (2+0), Organic & Inorganic Chemistry (2+1).
- **2nd Semester:** Applied Physics (3+1), Computer Aided Engineering Drawing (0+1), Chemical Process Calculations-I (3+0), Linear Algebra and Differential Equations (3+0), Reading & Writing Skills (3+0), Physical & Analytical Chemistry (2+1), Workshop Practice (0+1).
- **3rd Semester:** Programming Fundamentals (2+1), Electrical Technology and Electronics (2+0), Chemical Engineering Thermodynamics-I (3+1), Particle Technology (3+1), Fluid Mechanics-I (3+1).
- **4th Semester:** Chemical Process Calculations-II (3+0), Applied Mathematics in Chemical Engineering (3+0), Communication and Presentation Skills (3+0), Chemical Process Technology (2+1), Chemical Engineering Thermodynamics-II (3+0), Mass Transfer (3+0).
- **5th Semester:** Separation Processes (2+1), Fluid Mechanics-II (3+1), Numerical Analysis & Software Applications (2+1), Probability and Statistics (2+0), Heat Transfer (3+1), Engineering Mechanics (2+0).
- **6th Semester:** Fuels and Energy (3+1), Chemical Reaction Engineering (3+1), Simultaneous Heat and Mass Transfer (2+1), Engineering Materials (3+0), Technical Report Writing (2+0), Engineering Economics (2+0).
- **7th Semester:** Logical and Critical Thinking (2+0), Chemical Plant Design (3+0), Chemical Process Design and Simulation (2+1), Project-I (0+2), Transport Phenomena (3+0), Instrumentation and Process Control (3+1).
- **8th Semester:** Industrial Management (3+0), Elective-I (2+0), Elective-II (2+0), Entrepreneurship (2+0), Maintenance Engineering and Risk Management (2+0), Chemical Engineering Plant Design Project (0+4).

Track B: For Student Intake Terms Fall-2022 Onwards

- **1st Semester:** Islamic Studies/Ethics (2+0), Reading & Writing Skills (3+0), Calculus and

Analytical Geometry (3+0), Physical and Analytical Chemistry (2+1), Introduction to Computer and Programming Concepts (2+1), Introduction to Chemical Engineering (2+0), Engineering Drawing & Graphics (1+1).

- **2nd Semester:** Pakistan Studies (2+0), Applied Physics (3+1), Linear Algebra and Differential Equation (3+0), Chemical Process Calculations-I (2+0), Chemical Process Industries (2+0), Organic and Inorganic Chemistry (2+1), Workshop Practice (0+1).
- **3rd Semester:** Particle Technology (3+1), Chemical Engineering Thermodynamics (3+1), Communication and Presentation Skills (3+0), Applied Mathematics in Chemical Engineering (3+0), Fluid Mechanics-I (3+1).
- **4th Semester:** Numerical Methods for Chemical Engineers (2+1), Chemical Process Calculations-II (3+0), Logical and Critical Thinking (2+0), Electrical Technology and Electronics (2+0), Engineering Mechanics (2+0), Heat Transfer-I (3+1).
- **5th Semester:** Engineering Material (3+0), Fluid Mechanics-II (3+1), Heat Transfer-II (3+1), Mass Transfer (3+1), Probability and Statistics (2+0).
- **6th Semester:** Technical Report Writing (2+0), Engineering Economics (2+0), Simultaneous Heat and Mass Transfer (3+1), Chemical Reaction Engineering (3+1), Fuels and Energy (3+1), Maintenance Engineering and Risk Management (2+0).
- **7th Semester:** Instrumentation and Process Control (3+1), Chemical Plant Design (3+0), Elective-I (3+0), Process Modeling and Simulation (2+1), Project-I (0+2).
- **8th Semester:** Transport Phenomena (3+0), Elective-II (3+0), Biochemical Engineering (2+0), Industrial Management (3+0), Entrepreneurship (2+0), Project-II (0+4).

2. Master of Science in Chemical Engineering (MS Chemical Engineering)

Admission Requirements

- **Academic Background:** 16 years of education or equivalent (e.g., 4-year B.E./BS degree) in Chemical Engineering or Petroleum & Gas Engineering from an HEC-recognized University.
- **Academic Minimums:** At least **60% marks** under the annual grading setup, or a program CGPA of **2.5 out of 4.0**.
- **Standardized Testing:** A minimum score of **50% marks** on the GAT General assessment test.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses Listed:** 42 – 44 (*Note: This standard catalog total traditionally refers to baseline credit options, typical graduate tracks complete 8 to 10 actual semester courses*).
- **Minimum Academic Standing:** CGPA ≥ 2.5